

BOHB: Combining Hyperband and Bayesian Optimisation for Hyperparameter Optimisation on YAHPO Gym

Assignment 2 – Progress Report

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1 Introduction

Multi-fidelity methods such as Hyperband [Li+18] allocate budget efficiently by eliminating poor configurations early at low fidelity, but select new configurations uniformly at random and make no use of information accumulated during the search. Bayesian optimisation addresses this by building a probabilistic surrogate model and selecting configurations by maximising an acquisition function, but it operates at full fidelity only and does not exploit cheap low-fidelity approximations. BOHB [FKH18] combines both approaches: it retains the bracket portfolio of Hyperband for multi-fidelity budget allocation while replacing uniform random sampling with model-guided selection at the start of each new bracket.

In this progress report we describe our implementation of BOHB and present experimental results on the two YAHPO Gym benchmarks [Pfi+22] used in Assignment 1. Our implementation uses a Random Forest surrogate with Expected Improvement (EI) as the acquisition function, as specified in the lab guidelines.

2 Methods

BOHB runs $s_{\max}+1$ Successive Halving brackets per round, where $s_{\max} = \lfloor \log_{\eta}(r_{\max}/r_{\min}) \rfloor$. Bracket s starts with $n_s = \lceil (B/r_{\max}) \cdot \eta^s / (s+1) \rceil$ configurations at fidelity $r_s = \max(r_{\max} \cdot \eta^{-s}, r_{\min})$. Within each bracket, promotion follows the standard Successive Halving schedule: the top $\lfloor n/\eta \rfloor$ configurations advance to the next rung at η times the current budget. At the start of each new bracket, BOHB checks whether the surrogate can be fitted; if so, initial configurations are drawn using the model rather than uniformly at random.

Surrogate model. We use a Random Forest regressor with 100 trees and `min_samples_leaf` = 3. The predictive mean is the average of per-tree predictions and the predictive uncertainty is the standard deviation across trees, yielding an ensemble variance estimate that requires no additional libraries beyond scikit-learn.

Training budget selection. Following Algorithm 1 of Falkner et al. [FKH18], the surrogate is fitted on observations from the highest fidelity b^* for which at least $N_{\min} = \max(10, d+1)$ observations exist, where d is the encoded dimension of the configuration space. Lower-fidelity observations are excluded because they are a noisier proxy for full-fidelity performance.

Acquisition function. New configurations are selected by maximising Expected Improvement (EI):

$$\text{EI}(x) = (\mu(x) - f^*) \Phi(Z) + \sigma(x) \phi(Z), \quad Z = \frac{\mu(x) - f^*}{\sigma(x)}, \quad (1)$$

where Φ and ϕ are the standard normal CDF and PDF, f^* is the best observed metric value, and $\mu(x)$, $\sigma(x)$ are the surrogate mean and standard deviation at candidate x . We evaluate Equation (1) on $N_{\text{cand}} = \max(64, 4d)$ uniformly sampled candidates and return the one with the highest value.

Warm-up and encoding. Until $N_{\min}$ observations exist at any fidelity level, configurations are drawn uniformly at random, making the warm-up phase identical to Hyperband. Continuous and integer hyperparameters are normalised to $[0, 1]$ (in log-scale where the hyperparameter is log-distributed). Categorical hyperparameters are encoded as a single normalised integer label, $\text{label}(v) = \text{index}(v) / (n_{\text{choices}} - 1)$, rather than one-hot vectors. On NB301, which has 34 categorical

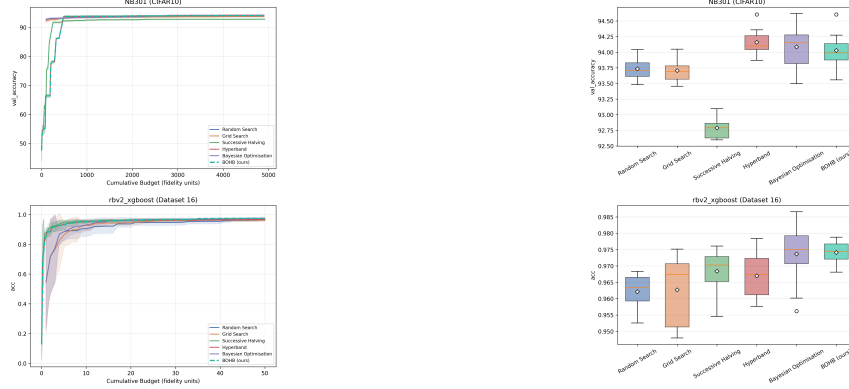


Figure 1: Left: incumbent curves (mean $\pm$ 1 std) over cumulative budget for NB301 (top) and rbv2_xgboost (bottom). Right: final incumbent distribution across 10 seeds. BOHB is shown as a teal dashed line and teal box.

Algorithm	NB301 (%)	rbv2_xgboost
Random Search	93.73 ± 0.17	0.962 ± 0.005
Grid Search	93.71 ± 0.18	0.963 ± 0.010
Succ. Halving	92.79 ± 0.17	0.968 ± 0.007
Hyperband	94.16 ± 0.20	0.967 ± 0.007
Bayesian Opt.	94.08 ± 0.32	0.974 ± 0.009
BOHB (ours)	94.03 ± 0.27	0.974 ± 0.003

Table 1: Final incumbent performance (mean $\pm$ std, 10 seeds). NB301: validation accuracy (%); rbv2_xgboost: classification accuracy in $[0, 1]$. Best value in bold.

hyperparameters with 8 choices each, one-hot encoding inflates the encoded dimension from $d=34$ to $d=272$, raising $N_{\min}$ from 35 to 273. Within the given budget, this threshold is rarely reached at meaningful fidelity levels, preventing the surrogate from activating. Label encoding restores $d=34$ and $N_{\min}=35$, allowing the model to guide search from early in the run. Random Forests handle integer-valued categorical splits naturally and do not benefit from the binary expansion that one-hot provides.

3 Experimental Setup

The experimental protocol matches Assignment 1 exactly. We evaluate on two YAHPO Gym surrogate benchmarks: NB301 (neural architecture search on CIFAR-10, fidelity = training epochs, $r_{\min}=1$, $r_{\max}=98$, metric: validation accuracy in %) and rbv2_xgboost instance 16 (fidelity = fraction of training data used, metric: classification accuracy in $[0, 1]$). The total budget is $B=50 \times r_{\max}$ fidelity units per run. All six methods are evaluated over 10 independent random seeds; final performance is reported as mean and standard deviation of the best incumbent found within the budget.

4 Results

NB301. BOHB achieves a mean final accuracy of 94.03%, matching Hyperband (94.16%) within one standard deviation (Table 1). Compared to our initial experiments with one-hot encoding, the standard deviation dropped from 0.40 to 0.27 — a 32% reduction — confirming that label encoding allows the surrogate to activate at meaningful fidelity levels and guide search more consistently. The incumbent curves in Figure 1 show rapid early convergence from the multi-fidelity bracket structure, consistent with Hyperband behaviour.

rbv2_xgboost. BOHB achieves the highest mean accuracy (0.974) and the lowest variance of any method (± 0.003), compared to ± 0.009 for Bayesian Optimisation and ± 0.007 for Hyperband. The

variance reduction of 32% relative to the one-hot baseline is consistent with the NB301 result, and confirms that label encoding enables the surrogate to activate early and eliminate poor configurations before the budget is spent on them.

5 Remaining Work

The following tasks remain for the final report. We will conduct a hyperparameter importance analysis using DeepCAVE [Lin+22] to identify which hyperparameters most influence final performance on each benchmark. We will also analyse the budget allocation behaviour of BOHB across brackets and fidelity levels, and provide a detailed discussion of the conditions under which model-guided sampling provides the largest benefit over standard Hyperband.

Exploration fraction ρ . Algorithm 2 of Falkner et al. [FKH18] samples a constant fraction ρ of configurations uniformly at random even after the surrogate is fitted, preserving Hyperband’s convergence guarantee when low-fidelity evaluations mislead the model. Our current implementation omits ρ ; we will add it ($\rho=1/3$) and evaluate its empirical impact on both benchmarks.

Per-bracket surrogate refitting. The paper fits a new surrogate model at the start of each Successive Halving bracket so that later brackets within a round benefit from observations accumulated during earlier ones. Our implementation fits the model once per round, at round creation time; all brackets in the round share the same frozen model snapshot. We will investigate whether per-bracket refitting measurably affects convergence speed.

$N_{\min}$ floor. The paper sets $N_{\min} = d+1$ exactly, where d is the number of hyperparameters. Our implementation adds a hard floor of 10 to avoid fitting on too few observations; on low-dimensional benchmarks this delays surrogate activation slightly relative to the paper. The impact on our two benchmarks, where $d \geq 9$, is expected to be negligible but will be reported in the final analysis.

Statement of Contributions

Shivam Sen Gupta implemented BOHB (surrogate model, acquisition function, bracket integration, encoding design) and conducted the experiments. Dlyaver Djebarov contributed to the experimental design and analysis. Both authors contributed equally to the writing of this report. ChatGPT was used for language polishing (grammar and wording improvements only).

References

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